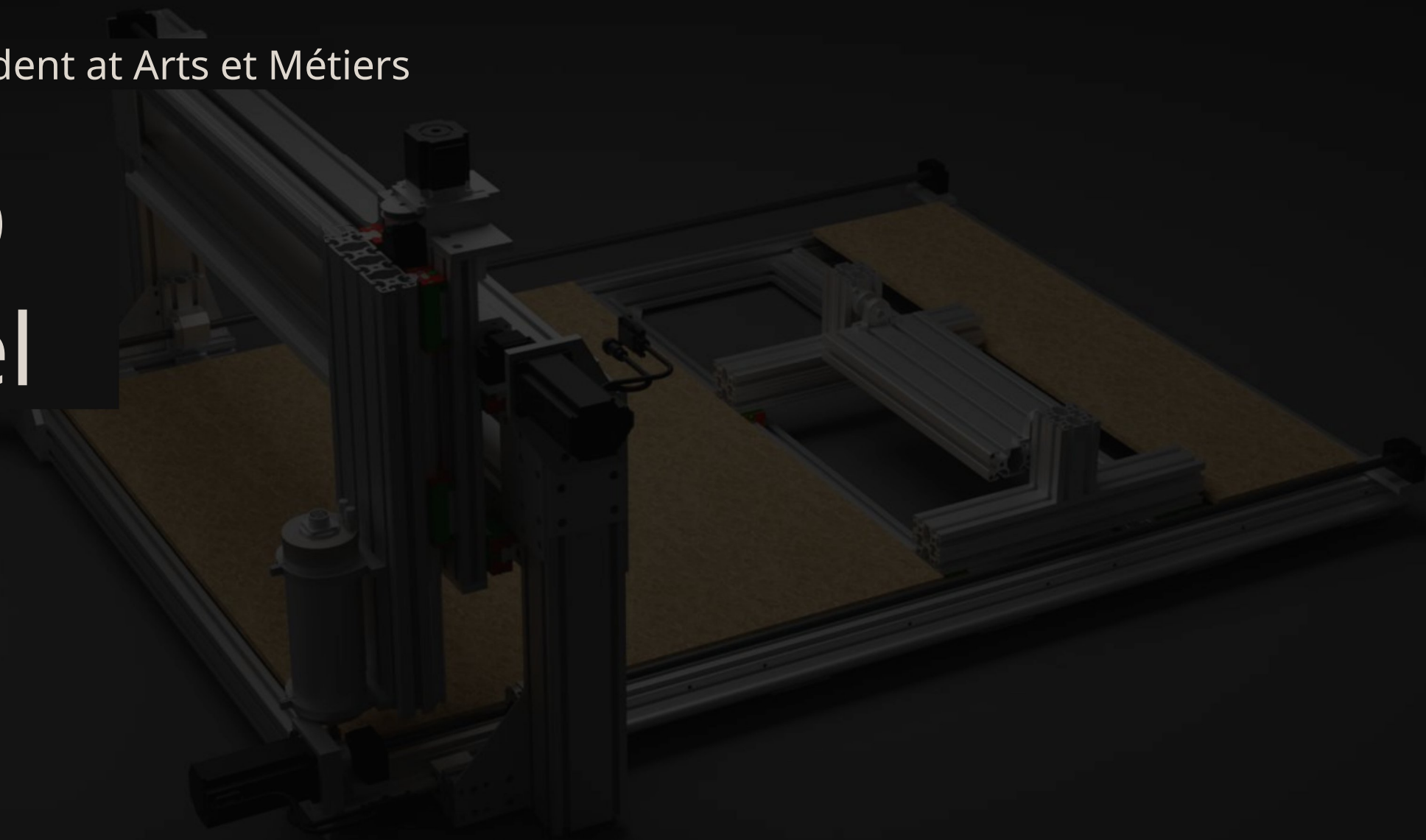


Engineering student at Arts et Métiers

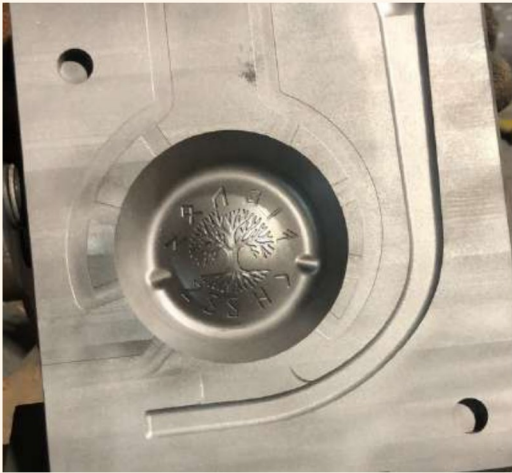
DIARD Samuel

PORTFOLIO



Projects

**INDUSTRIALISATION -
METAL MOULDS AND**



**SERIES PRODUCTION
JEWELLERY -
LOST-WAX CASTING**



**THERMAL ENGINEERING -
PROGRAMMABLE FURNACE
1200 °C**



**PRECISION MACHINERY
- 5-AXIS CNC MILLING MACHINE**



Mass production – Metal moulds and mass production

BACKGROUND:

For over half a century, the creation of a commemorative ashtray has been a tradition at the Grand Gala des Fignos des Arts et Métiers. For the 2024 edition, a new design was created in keeping with the event's theme (The Vikings), with the aim of continuing this tradition through the production of 300 aluminium pieces. This project, for which I was responsible, involved developing the entire manufacturing process, from the design of the ashtray through to the creation of the moulds and the production of the parts.

OBJECTIVES:

To design and implement a manufacturing process enabling the mass production of these ashtrays, from the design of the component through to their final production.

The project involved:

- the design of the ashtray;
- the development of the casting system;
- the design and machining of the cast iron moulds;
- the casting of the aluminium components.



One of the two mould halves



An ashtray after casting and finishing

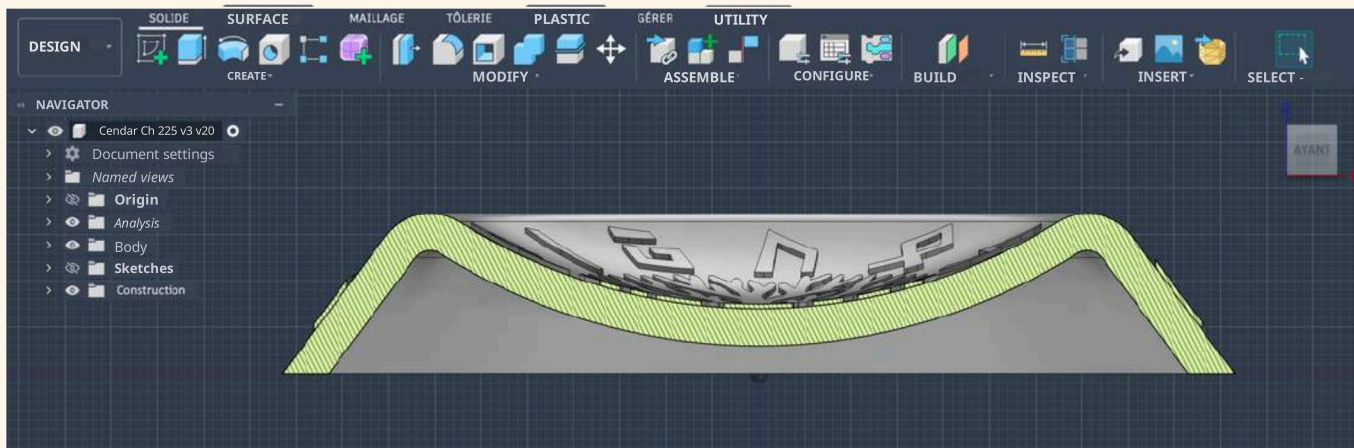
Industrialisation – Metal mould and mass production

DESIGN OF THE ASHTRAY:

Design incorporating the constraints of shell moulding right from the CAD stage: no undercuts, sloping walls for demoulding, and constant wall thickness to prevent shrinkage.

Particular attention was paid to the design for demoulding: undercuts were avoided, vertical walls were angled to facilitate extraction, and a constant wall thickness was prioritised to minimise the risk of shrinkage during solidification.

The patterns and inscriptions were incorporated directly into the model with a material allowance, creating a recessed relief in the mould and simplifying their machining.



The 3D model of the ashtray, shown in cross-section and from above. CAD design created in FUSION 360.

CAD model of the gating system, its technical drawing and one of the two mould halves

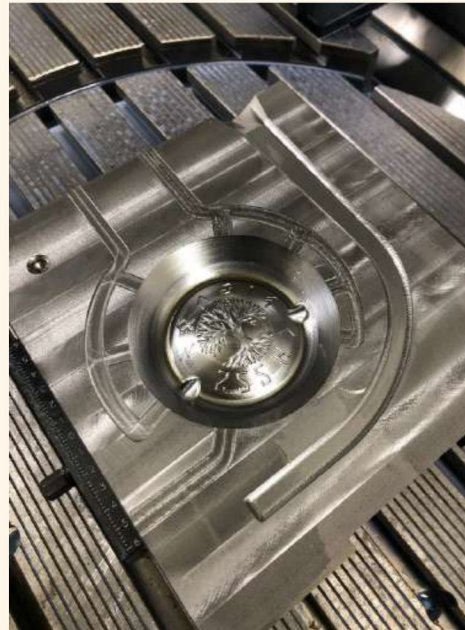
Industrialisation – Metal mould and mass production

MOULD MANUFACTURE:

EN-GJL 250 cast iron blocks, ground and then CNC-machined based on the CAD model. Tapping, sandblasting and coating complete the tooling prior to the first casting, to ensure precision and clean demoulding.

The two mould halves were produced from EN-GJL 250 lamellar cast iron blocks (250 × 250 × 40 mm). The bearing surfaces were first ground on a conventional milling machine to achieve a precise reference geometry, before the cavities were machined on a CNC milling machine based on the CAD model. Additional operations incorporated the elements required for the mould's use: M14 tapped holes for securing the handles and retaining flanges, preparation of the functional surfaces, and adjustment of the assembly components.

Before the first casting, the moulds were sandblasted and then annealed to remove any cutting oil residues absorbed by the cast iron during machining. Finally, a release coating was applied to the surfaces in contact with the aluminium to protect the mould, facilitate demoulding and improve the surface quality of the parts.



Mould halves fresh from machining and finalised production tooling

Industrialisation – Metal mould and mass production

PART PRODUCTION:

300 ashtrays cast at 700 °C in moulds preheated to 350 °C, at a rate of approximately one part per minute. The feeding system was designed to allow removal without sawing, thereby reducing finishing time.

The moulds were preheated to approximately 350 °C to minimise thermal shock, ensure uniform filling of the cavity and improve the quality of the resulting parts. The aluminium was then cast at 700 °C directly into the mould, enabling a production rate of approximately one part per minute.

The mould was also designed with a view to optimising subsequent operations: the gates were positioned and dimensioned to allow for easy removal of the feeding system after solidification, without the need for sawing, as the sprue can be separated from the part by simple mechanical breaking, thereby reducing finishing time.



Removal of the gating and final result after deburring

Industrialisation – Metal mould and mass production

CONCLUSION:

This project resulted in the production of a complete series of 300 aluminium ashtrays by developing a fully controlled manufacturing process, from the design of the component through to the production of the finished items. The design of the tooling and the casting system enabled repeatable production from the very first trials, whilst minimising finishing operations. This project enabled me to develop a holistic approach to product development, by incorporating manufacturing constraints right from the design stage. In particular, it taught me the importance of anticipating the production stages in order to design not only a functional part, but also a reliable and reproducible process.

KEY FIGURES:

- 300 parts produced
- Over 40 hours of CNC and conventional machining
- 50 kg of aluminium
- 35 kg of cast iron
- 60 seconds' cycle time

From digital design to the finished part, this project illustrates my commitment to developing solutions where technical expertise and an understanding of manufacturing processes are harnessed to achieve a tangible result.

SKILLS DEVELOPED:

Mechanical design:

- Design of a component for a foundry process
- Design of production tooling
- Incorporation of manufacturing constraints from the CAD stage onwards

Manufacturing processes:

- CNC machining of cast iron moulds
- Aluminium shell moulding
- Development of a production process

Project methodology:

- Management of a complete development project
- Resolving technical issues
- Transition from prototype to small-batch production

Jewellery – Lost-wax casting

BACKGROUND:

This ring was created to mark my parents' 25th wedding anniversary. They provided me with the stone – a 7.79-carat synthetic blue diamond – with the intention of having it set in a unique piece, designed and crafted entirely by me. The result is a flower-shaped silver setting, designed to showcase the stone whilst ensuring its structural integrity.

OBJECTIVES:

The design was developed based on my mother's expectations, incorporating the standard constraints of lost-wax casting and stone-setting from the very first sketches – the castability of the model and the mechanical strength of the setting to support a 7.79-carat stone.

The project covered the entire production process: CAD design, development of the casting system, printing of the wax model, lost-wax casting in silver, polishing and final setting of the stone.



Finished ring and 3D rendering

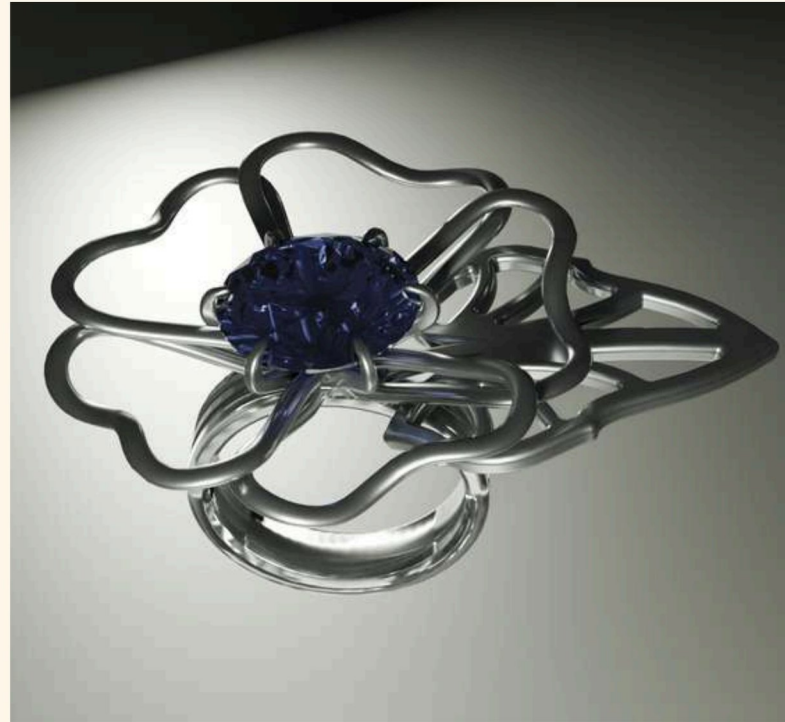
Jewellery – Lost-wax casting

CAD AND 3D PRINTING:

3D model created using a combination of Fusion 360 and Blender, prior to integrating the casting system. The ring was then 3D-printed in wax using an investable resin, ready for casting.

Once the shape of the ring had been finalised, the 3D model was created using both Fusion 360 and Blender: the former for geometric precision and mechanical constraints, the latter to refine the organic shapes of the flower. The casting system (feed channel, positioning) was then integrated directly into the model.

The ring was finally printed in casting resin (PAC10 HeyGears) using a 3D printer, producing a 'wax' model ready for casting.



Blender rendering of the CAD model and the wax part fresh from the 3D printer

Jewellery – Lost-wax casting

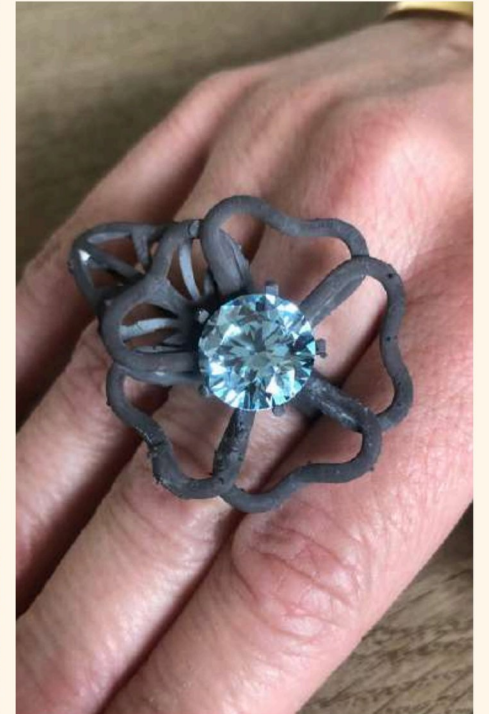
CASTING OF THE ROUGH CAST:

The wax part is embedded in a casting cylinder filled with degassed foundry plaster, then fired for approximately 12 hours at up to 730 °C to remove the wax. Molten silver (1000 °C) is poured under vacuum at 550 °C, followed by demoulding by breaking the plaster.

The 3D-printed wax model is placed in a steel casting cylinder, into which foundry plaster is poured. This plaster is degassed before and after casting to remove any air bubbles that could cause defects in the final piece.

The cylinder is then fired for around twelve hours, at a maximum temperature of 730 °C, to completely remove the wax and fire the plaster. The temperature is lowered to 550 °C before the molten silver (925 silver), at 1000 °C, is poured under vacuum to ensure the mould is completely filled.

Once the piece has solidified, the plaster is broken to release the rough cast.



Refractory plaster in the cylinder, metal casting and the rough-cast piece with its stone on top

Jewellery – Lost-wax casting

POLISHING AND SETTING:

Acid etching, removal of casting flash, followed by gradual polishing to a mirror finish (using sandpaper, a brush, felt and red/black polishing pastes, with ultrasonic treatment between each stage). Claw setting, the length of which was determined at the casting stage: all that remains is to fold them over the stone.

The rough-cast piece is first treated with acid to remove the oxide formed during casting, then the casting flash is removed using pliers and a saw. Rough sanding with sandpaper removes the main surface irregularities, followed by cleaning with a brush fitted to a Dremel.

The piece is then polished with felt using polishing paste – first red, then black for the mirror finish – with ultrasonic cleaning between each stage to remove any paste residue and visually check the progress.

Finally, the stone is secured using a claw setting, the length of which was planned during the design and casting of the setting. All that remains is to fold the claws down one by one over the stone, taking care not to apply excessive pressure given its size.



Ring being polished and the final result after setting

Jewellery – Lost-wax casting

CONCLUSION:

This project enabled me to create a unique piece of jewellery from start to finish, from the 3D design right through to the final setting, whilst incorporating, from the design stage, the specific constraints of the lost-wax casting process and the need to showcase a stone of considerable size. Beyond the technical aspects, this ring combines precision craftsmanship with high aesthetic standards – two central requirements in my approach to jewellery-making.

This project also gave me the opportunity to work for a client with specific expectations – in this case, my mother – and to deliver a piece that met both a strong aesthetic demand and demanding technical constraints.

KEY FIGURES:

- 7.79 carats (Fancy Vivid Blue Diamond)
- 13 g (925 sterling silver)

Creating this ring enabled me to develop a comprehensive approach to jewellery-making, from digital design through to hand finishing. In particular, this project taught me the importance of anticipating, right from the CAD stage, the manufacturing constraints specific to lost-wax casting and stone-setting, in order to produce a piece that is both aesthetically pleasing and technically flawless.

SKILLS DEVELOPED:

Design

- Mixed CAD modelling (Fusion 360 / Blender)
- Design incorporating castability and setting constraints

Manufacturing process

- 3D printing with castable resin
- Silver lost-wax casting

Jewellery

- Polishing
- Claw setting

Thermal engineering – Programmable 1200°C furnace

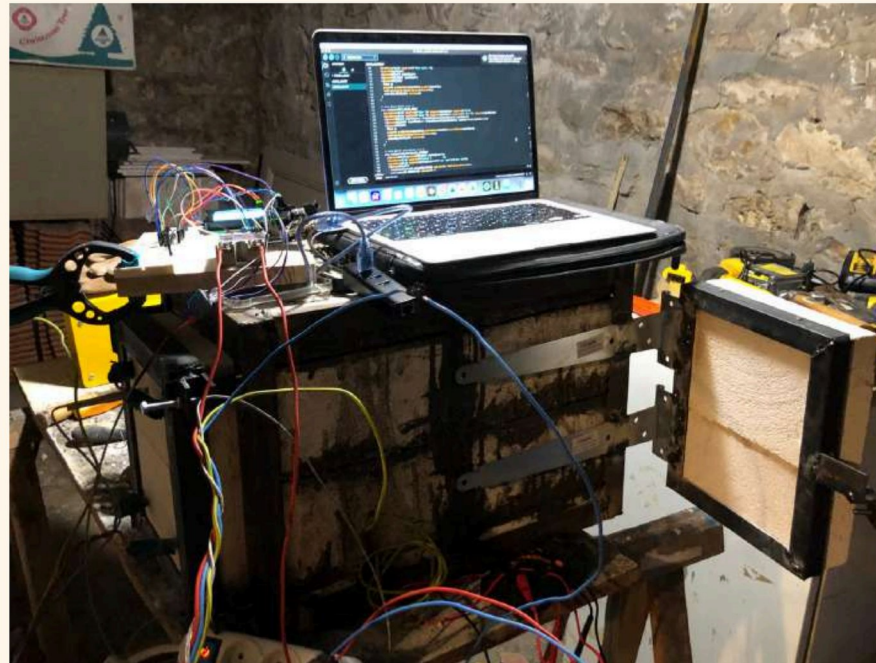
BACKGROUND:

The production of forged and lost-wax cast components requires precise control of thermal cycles. As part of my personal projects, I have developed a programmable electric furnace capable of meeting two distinct requirements: forging work, particularly for cutlery, which requires a precise heating temperature to be maintained; and the firing of plaster moulds used in lost-wax casting, which necessitates the automatic monitoring of a temperature curve in order to gradually remove the wax without damaging the mould.

OBJECTIVES:

To design an electric furnace capable of reaching 1200°C with stable temperature control and versatile use. The aim was to produce a reliable, safe tool suitable for various manufacturing processes thanks to its two operating modes.

- Hold mode: stable temperature for forging or heat treatment
- Programme mode: automatic tracking of a heating curve for processes requiring precise control



Furnace and its electronics under development, and the forging of a knife

Thermal engineering – 1200°C programmable furnace

ARCHITECTURE:

Welded steel frame, class 26 refractory bricks and ceramic fibre for insulation. Heating provided by bespoke Kanthal A1 heating elements (3 kW), controlled by an Arduino via SSRs and a type K thermocouple.

The furnace combines Class 26 refractory bricks, selected for their ability to withstand temperatures up to 1430 °C and their balance of mechanical strength and thermal insulation. They are assembled using refractory mortar within a welded steel frame, then covered with ceramic fibre to minimise heat loss and improve energy efficiency.

Heat is generated by bespoke Kanthal A1 heating elements (3 kW), controlled by a solid-state relay (SSR) modulated by an Arduino UNO, combined with a type K temperature probe for precise monitoring.



Machining of the grooves for the heating elements and assembly of the furnace with its door

Thermal engineering – Programmable 1200°C furnace

MANUFACTURE OF HEATING ELEMENTS:

Helical Kanthal A1 heating elements rated for 3 kW at 230 V (13 A, 18 Ω), then custom-twisted from resistance wire.

The aim was to design helical heating elements made from Kanthal A1, a ferritic iron-chromium-aluminium alloy renowned for its mechanical strength and resistance to oxidation at high temperatures, to deliver a total power of 3 kW.

To achieve 3 kW at 230 V, a current of approximately 13 A must flow through the heating element, resulting in a target electrical resistance of approximately 18 Ω . The required length of wire was then determined based on the resistivity of Kanthal A1 and the chosen cross-section – a compromise between mechanical strength at high temperatures, overall dimensions and manufacturing cost.

The heating element was made from twisted Kanthal wire using a wooden template and a spindle attached to a screwdriver.



Resistance wire, resistance element shaped and in operation in the furnace

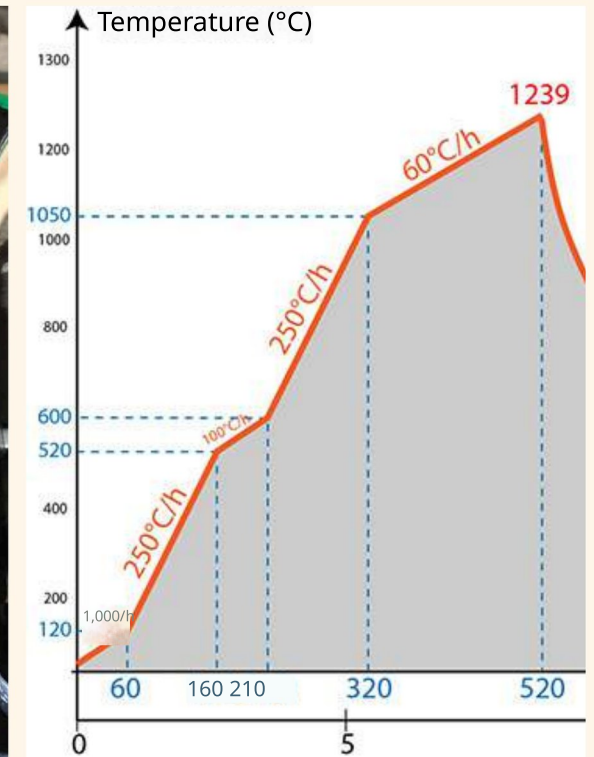
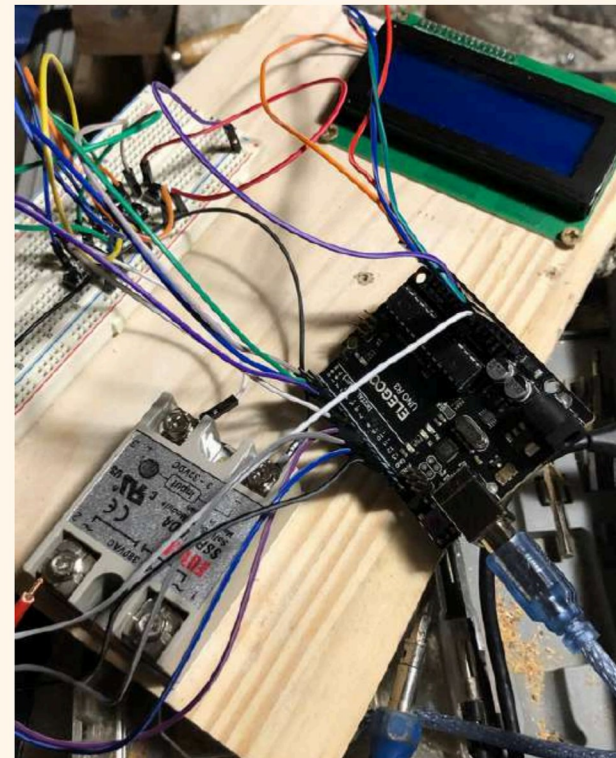
Thermal engineering – Programmable 1200°C oven

ELECTRONICS IMPLEMENTATION:

Control via pulse width modulation: an SSR driven by an Arduino UNO, alternating between full power and cut-off according to the setpoint. The furnace's thermal inertia smooths out these variations, ensuring stable control, whether maintaining a fixed temperature or following a programmed curve.

To regulate the temperature precisely, two approaches were possible: a continuous variation in the voltage across the resistors – which is complex at this power level – or control via pulse width modulation. The latter solution was chosen: the furnace alternates between phases at maximum power and phases with no power supply, with the thermal inertia of the system naturally smoothing out these rapid variations.

Switching is handled by a solid-state relay (SSR), which was preferred to an electromechanical contactor due to its durability, lack of mechanical wear and ability to withstand a large number of cycles. The SSR is controlled by an Arduino UNO board receiving data from a type K temperature probe (measuring up to $\approx 1250\text{ }^{\circ}\text{C}$), enabling it to maintain a fixed setpoint or automatically follow pre-recorded firing programmes, particularly for lost-wax cycles.



Assembled electronics and example of a firing curve

Thermal engineering – Programmable furnace 1200°C

CONCLUSION:

This project resulted in the design and construction of a programmable electric furnace capable of meeting a range of requirements related to thermal processes. The integration of electronic control now makes it possible to maintain a set temperature or automatically follow complex firing cycles, particularly for the production of refractory plaster moulds used in the manufacture of foundry components using the lost-wax casting technique.

KEY FIGURES:

- 1200°C
- 3,000 W
- 38 m of heating elements
- over 600 lines of code

Building this furnace enabled me to develop a holistic approach to the design of industrial equipment, taking into account mechanical, thermal and electronic constraints simultaneously. In particular, this project taught me the importance of component sizing and the interaction between different subsystems when designing a machine.

SKILLS DEVELOPED:

Mechanical design

- Design of a welded structure
- Selection of high-temperature materials
- Design of a refractory assembly

Thermal & Process Engineering

- Sizing a heating system
- Management of thermal cycles

Electronics & automation

- Power control using SSRs
- Temperature acquisition using thermocouples
- Arduino programming and thermal control

Precision engineering – 5-axis CNC milling machine

BACKGROUND:

I had previously built my first 3-axis CNC milling machine using a design I found online. This project allowed me to familiarise myself with the mechanical design of this type of machine, as well as its rigidity and precision requirements. The machine was eventually sold, which inspired me to take things further with a new project.

Building on this initial experience, this time I wanted to design a machine entirely from scratch, without using an existing plan, and add two extra axes to upgrade from a 3-axis to a 5-axis configuration.

OBJECTIVES:

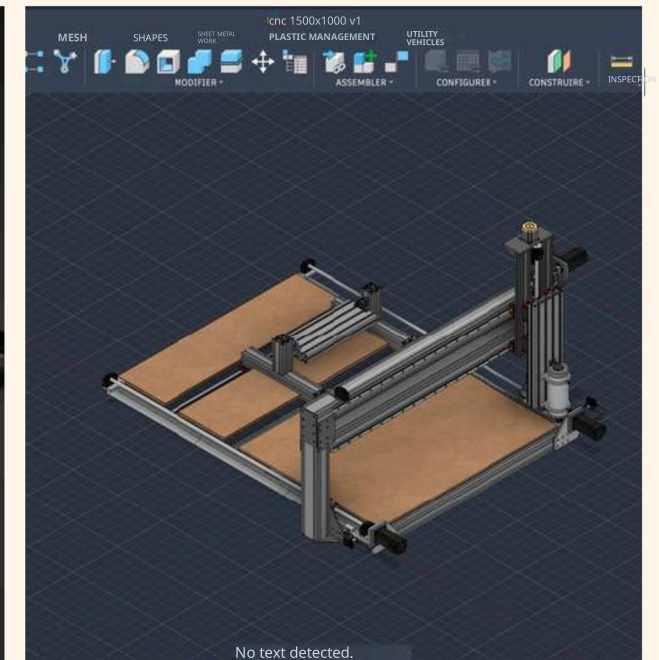
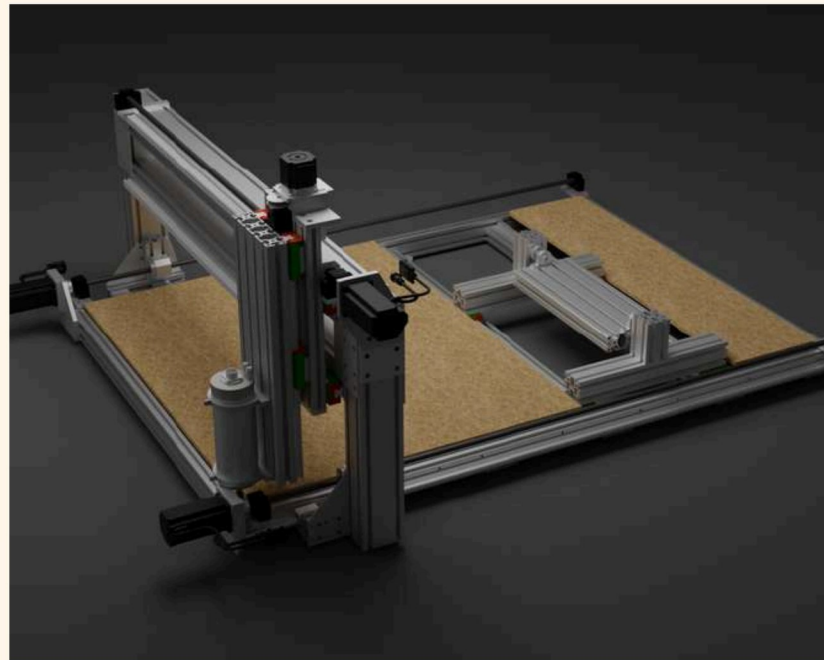
To design and build a CNC milling machine capable of machining parts using multi-axis operations.

To design a mechanical architecture allowing for the simultaneous positioning of 5 axes.

To dimension and integrate the various sub-assemblies: frame, guides, ball screws, spindle and motorisation.

Develop the control electronics and integrate the various axes.

Find a balance between rigidity, precision, footprint and cost.



3D rendering and CAD

Precision engineering – 5-axis CNC milling machine

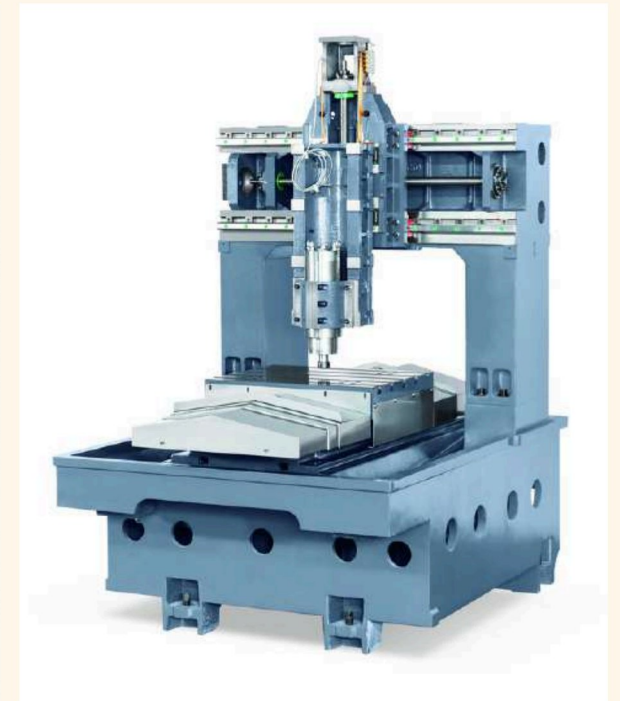
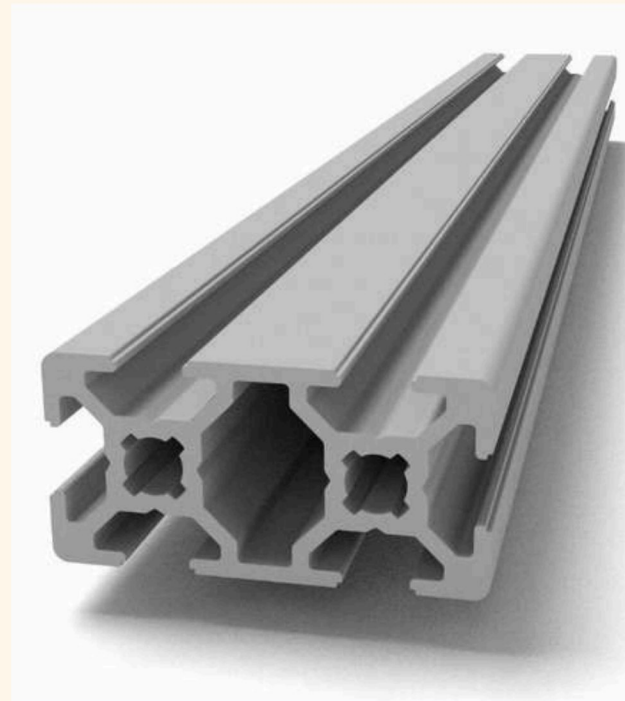
TECHNICAL SOLUTIONS:

Several options were considered for the structure: welded steel, epoxy-granite or bolted aluminium sections. The choice fell on the latter solution, offering the best balance between rigidity, ease of manufacture, weight and the ability to produce a large-volume machine.

The machine's structure is made from aluminium sections assembled using bolts. This solution allows for rapid manufacture and offers relatively simple alignment, whilst maintaining sufficient rigidity for the required machining volume.

A welded steel structure would have provided greater rigidity, but the distortions caused by welding would have made aligning the guides more complex. Conversely, an epoxy-granite frame would have offered excellent rigidity and good vibration damping, but at the cost of a more complex construction, significant mass and poorer suitability for a large-volume machine.

The choice of aluminium profiles is therefore a compromise between rigidity, precision, ease of manufacture, mass and machining volume.



Aluminium profile and example of a steel frame on a CNC machine

Precision engineering – 5-axis CNC milling machine

DESIGN:

The machine was designed in Fusion 360 using the CAD models of the components supplied by the manufacturers. The architecture is based on linear rails, ball screws and NEMA 34 closed-loop stepper motors, selected for their precision and reliability.

The design was carried out in Fusion 360, notably incorporating the CAD models of the aluminium profiles obtained from the supplier in order to design directly around the actual dimensions of the components.

The axes are guided by linear rails, offering good precision and a high load capacity. Power transmission is provided by ball screws, chosen for their precision and low friction, thereby minimising power losses.

The drive system utilises NEMA 34 closed-loop stepper motors. These offer a balance between the simplicity of conventional stepper motors and the performance – as well as the higher cost and complexity – of servo motors.



NEMA 34 motor, its driver, HGR20 linear rails and SFU1605 ball screws

Precision Machinery – 5-axis CNC milling machine

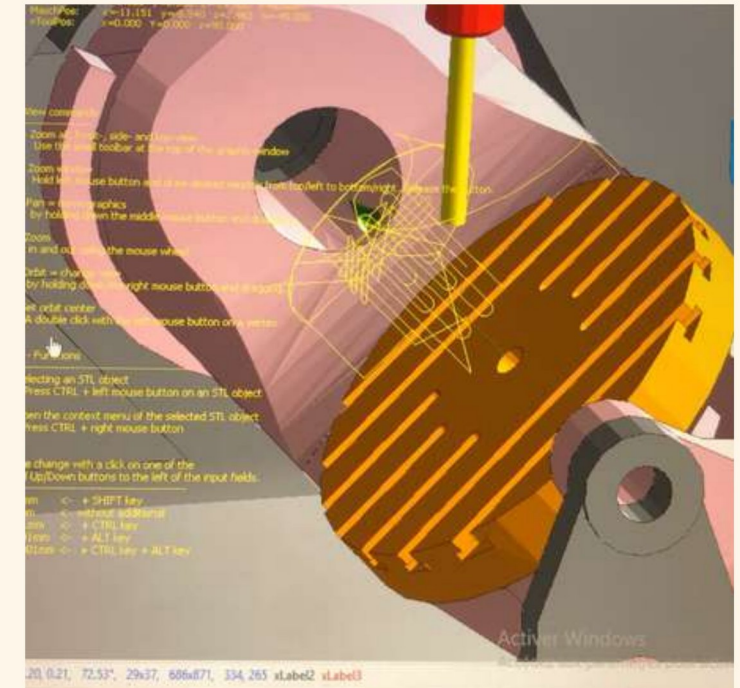
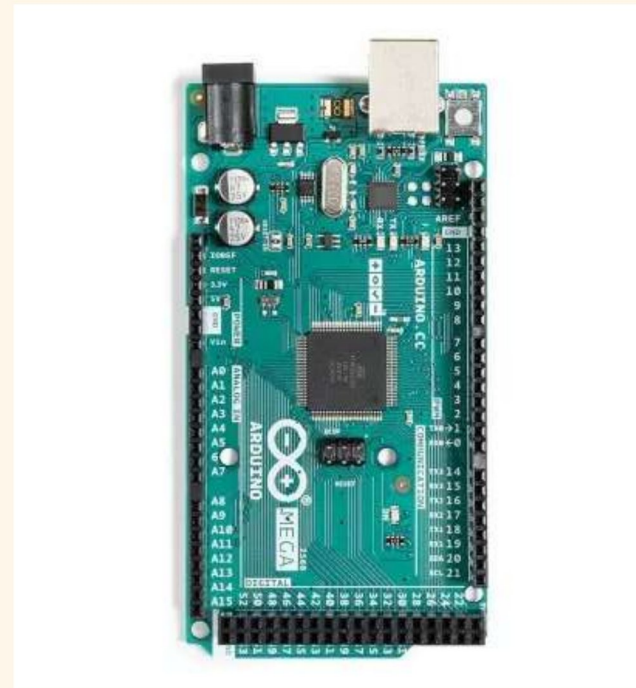
ELECTRICAL ARCHITECTURE:

The machine is controlled from a computer via GRBLGru, which communicates via a serial connection with an Arduino Mega fitted with a CNC Shield. The board runs GRBL Mega 5X to manage the five axes, whilst dedicated drivers control the motors in a closed-loop system. A 2.2 kW liquid-cooled spindle is also controlled by a frequency converter.

The machine is controlled from a computer using the GRBLGru software, which transmits instructions via a serial link to an Arduino Mega fitted with a CNC Shield. The board uses GRBL Mega 5X, an adaptation of GRBL that enables the management of five axes on an Arduino Mega.

The control signals are then transmitted to the motor drivers, which control the motors and manage the position feedback required for closed-loop operation.

Machining is carried out using a liquid-cooled spindle, driven by a 2.2 kW variable-frequency drive. This enables the spindle's rotational speed to be controlled according to machining requirements.



Arduino Mega board and machining simulation on GRBL Gru

Precision engineering – 5-axis CNC milling machine

MANUFACTURING:

The machine is currently under construction. The fixed frame and the moving gantry forming the Y-axis have already been completed, whilst some parts have been provisionally produced using 3D printing to enable assembly and initial testing.

The machine is currently being built. The fixed frame and the mobile gantry for the Y-axis have already been fabricated and assembled, allowing the various components to be integrated progressively.

Some parts have deliberately been produced on a temporary basis using 3D printing. These enable the assembly and operation of the machine to be validated before the final versions are manufactured from aluminium or steel.

Once the machine is sufficiently operational, the first final parts will be machined directly using it. The manufacturing process is therefore designed to be progressive: the machine gradually enables the production of the components required for its own completion.



Machine chassis being assembled

Precision engineering – 5-axis CNC milling machine

CONCLUSION:

This project enabled me to design a complete machine tool by incorporating the constraints of rigidity, precision, motorisation and control right from the initial stages. From the design of the mechanical architecture to the construction of the frame and the integration of the electronics, it led me to approach industrial design as a whole.

Still in the process of being built, this project also represents a step-by-step approach in which the machine itself is used to produce some of the components required for its completion. Above all, it has enabled me to develop a holistic approach to mechanical design, by continually reconciling the choices made in CAD with the practical constraints of manufacturing and assembly.

KEY FIGURES:

- 5 axes
- 2.2 kW – spindle power
- NEMA 34 – closed-loop motorisation
- 100 kg – weight of the machine once completed

Building this machine has enabled me to deepen my skills in mechanical design and manufacturing, whilst developing a more comprehensive understanding of mechatronic system integration. This project is still ongoing and represents a development programme that I am gradually continuing to work on.

SKILLS DEVELOPED:

Design

- Design of a complete machine tool
- Sizing and integration of mechanical components

Manufacturing

- CNC machining
- Assembly and alignment

Mechatronics

- 5-axis control
- Integration of motors, drivers and sensors
- Arduino Mega and GRBL Mega 5X

Other projects

The projects presented earlier represent a selection of the work that best defines my professional journey. Other projects—whether one-off assignments or initiatives in different fields—complement this experience and demonstrate the diversity of techniques I have employed.



3-axis CNC milling machine:
Construction of a 3-axis CNC wood milling machine based on plans.



Wood-carbon foils:
Design and manufacture of wood and carbon fiber foils for a radio-controlled sailboat.



Modular wind tunnel:
Resizing and 3D-printed fabrication of an experimental wind tunnel. Design of a 2-axis test rig to measure lift and drag.